

Example 20.1

Determine the value of the resistor, R , needed in Figs. 20.2 and 20.3 so that the reference drain currents are $20\text{ }\mu\text{A}$. Use the long-channel parameters seen in Table 9.1. Simulate the operation of the NMOS mirror with the calculated resistor value.

In Fig. 20.2 (the NMOS current mirror), we can write

$$I_{REF} = 20\text{ }\mu\text{A} = \frac{V_{DD} - V_{GS1}}{R} \approx \frac{KP_n}{2} \cdot \frac{10}{2} \cdot \left(\overbrace{\frac{1.05\text{ V}}{V_{GS1}}} - \overbrace{\frac{0.8}{V_{THN}}} \right)^2 = \frac{KP_n}{2} \cdot \frac{10}{2} \cdot (0.25)^2$$

(notice how we indicate “approximately” because channel-length modulation is not included in the equations) or

$$R = \frac{5 - 1.05}{20\text{ }\mu\text{A}} \approx 200\text{ k}\Omega$$

For Fig. 20.3 (the PMOS current mirror), we can write

$$I_{REF} = 20\text{ }\mu\text{A} = \frac{V_{DD} - V_{SG}}{R} \approx \frac{KP_p}{2} \cdot \frac{30}{2} \cdot \left(\overbrace{\frac{1.15\text{ V}}{V_{SG}}} - \overbrace{\frac{0.9}{V_{THP}}} \right)^2 = \frac{KP_p}{2} \cdot \frac{30}{2} \cdot (0.25)^2$$

or

$$R = \frac{5 - 1.15}{20\text{ }\mu\text{A}} \approx 200\text{ k}\Omega$$

Simulation of the operation of the NMOS current mirror is seen in Fig. 20.4. The reference current isn’t exactly $20\text{ }\mu\text{A}$ (and we shouldn’t expect it to be). The x-axis is a sweep of the voltage across the current source, V_O . Note that below $V_{DS,sat}$ ($= 250\text{ mV}$ here) M2 triodes and the output current, I_O , goes to zero. The output *compliance* range for this current source (the range of output voltages where the current source behaves like a current source, that is, not an open or a resistor) is between V_{DD} and $V_{DS,sat}$. The point where $V_O = V_{DS1} = V_{GS1}$ is where $I_O = I_{REF}$ (again this is important for matching two currents). Finally, note that I_{REF} and V_{GS1} are not dependent on V_O . ■

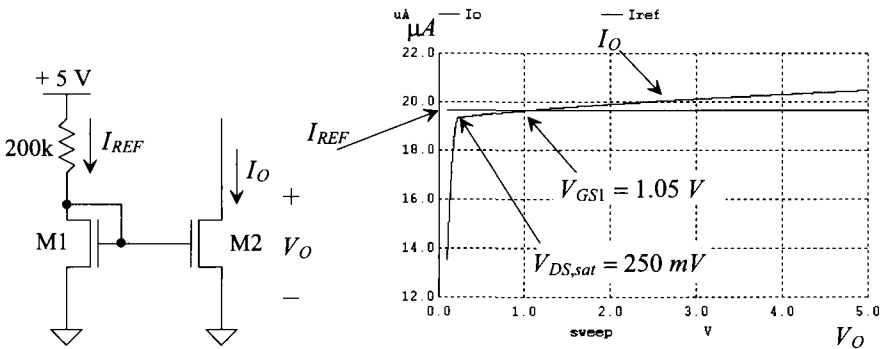


Figure 20.4 The operation of an NMOS current mirror.

20.1.2 Matching Currents in the Mirror

Many analog applications are susceptible to errors due to layout. In circuits in which devices need to be matched, layout becomes a critical factor. For example, in the basic current mirror shown in Fig. 20.2, first-order process errors can cause the output current, I_O , to be significantly different from the reference current. Process parameters such as gate-oxide thickness, lateral diffusion, oxide encroachment, and oxide charge density can drastically affect the performance of a device. Layout methods can be used to minimize the first-order effects of these parameter variations.

Threshold Voltage Mismatch

In a given current mirror application, the values for the threshold voltages are critical in determining the overall accuracy of the mirror. Again, examine the basic current mirror shown in Fig. 20.2. Since both devices have the same value for V_{GS} and assuming the sizes and transconductance parameters for both devices are equal, let's examine the effect of a mismatch in threshold voltages between the two devices. If it is assumed that the threshold mismatch is distributed across both devices such that

$$V_{THN1} = V_{THN} - \frac{\Delta V_{THN}}{2} \quad (20.5)$$

$$V_{THN2} = V_{THN} + \frac{\Delta V_{THN}}{2} \quad (20.6)$$

where V_{THN} is the average value of V_{THN1} and V_{THN2} and ΔV_{THN} is the mismatch, then

$$\frac{I_O}{I_{REF}} = \frac{\frac{KP_n W}{2 L} (V_{GS} - V_{THN} - \frac{\Delta V_{THN}}{2})^2}{\frac{KP_n W}{2 L} (V_{GS} - V_{THN} + \frac{\Delta V_{THN}}{2})^2} = \frac{\left[1 - \frac{\Delta V_{THN}}{2(V_{GS} - V_{THN})}\right]^2}{\left[1 + \frac{\Delta V_{THN}}{2(V_{GS} - V_{THN})}\right]^2} \quad (20.7)$$

If both expressions are squared and the higher order terms are ignored, then the first-order expression for the ratio of currents becomes

$$\frac{I_O}{I_{REF}} \approx 1 - \frac{2\Delta V_{THN}}{V_{GS} - V_{THN}} = 1 - \frac{2\Delta V_{THN}}{V_{DS,sat}} \quad (20.8)$$

Equation (20.8) is quite revealing because it shows that as V_{GS} decreases, the difference in the mirrored currents increases due to threshold voltage mismatch. This is particularly critical for devices that are separated by relatively long distances because the threshold voltage is susceptible to process gradients. *To attain high speed and to reduce the effects of threshold voltage mismatch, a large gate overdrive voltage should be used* (remembering for a long-channel process that $V_{ovn} = V_{DS,sat} = V_{GS} - V_{THN}$). Of course the drawback, for a current mirror, is a reduced range of compliance (the MOSFET enters the triode region earlier).

Transconductance Parameter Mismatch

The same analysis can be performed on the transconductance parameter, KP_n . If $KP_{n1} = KP_n - \Delta KP_n/2$ and $KP_{n2} = KP_n + \Delta KP_n/2$, where KP_n is the average of KP_{n1} and KP_{n2} , then assuming perfect matching on all other parameters, the difference in the currents becomes

$$\frac{I_O}{I_{REF}} = \frac{KP_n + 0.5\Delta KP_n}{KP_n - 0.5\Delta KP_n} \approx 1 + \frac{\Delta KP_n}{KP_n} \quad (20.9)$$

Since KP_n is a process parameter, we might think that as CMOS scales downwards (C'_{ox} goes up) the mismatch due to oxide variations and mobility differences might get better. However, there is less averaging of these variations with the smaller layout sizes. Practically, as we're about to see, differences in V_{DS} dominate the matching behavior.

Drain-to-Source Voltage and Lambda

One aspect of current mirror design that is *critical* for generating accurate currents is the drain-to-source voltage. As seen in Fig. 20.4, the only point where the currents of the devices are actually the same is when their V_{DS} values are equal. In Eq. (20.3) the ratio of the output current to the reference current is affected by both the matching in the drain-to-source voltages (V_O and V_{DS1}) and the device λ s. If, for example in the short-channel process (see Table 9.2), $V_{DS1} = .35$ V, $V_{DS2} = V_O = .75$ V, and $\lambda_1 = \lambda_2 = 0.6$ V⁻¹, then

$$\frac{I_O}{I_{REF}} = \frac{1 + \lambda_2 \cdot V_O}{1 + \lambda_1 \cdot V_{DS1}} = \frac{1 + 0.6 \cdot 0.75}{1 + 0.6 \cdot 0.35} = 1.20 \quad (20.10)$$

resulting in 20% error! It is extremely important, for good matching, that the V_{DS} values of the MOSFETs in the current mirror are equal.

Layout Techniques to Improve Matching

Most general analog applications require the length of the gate to be longer than minimum since the channel-length modulation, λ , has less effect on longer devices than on shorter ones (as discussed in Ch. 9). As a result, the minimum-sized devices found in digital circuits are not used as often in general analog design (however, see Ch. 26). However, the larger devices can result in larger parasitics if some layout issues are not considered. Figure 20.5a illustrates a basic MOSFET device with a large W/L . The implant resistance of the source and drain can be modeled as shown in Fig. 20.5b. The implant resistance can easily be reduced by simply adding as many contacts as possible along the width of both the source and the drain as seen in Fig. 20.5c. The increase in the number of contacts results in lower resistance, more current capability, and a more distributed current load throughout the device. However, as the device width increases, another technique is used which distributes the parasitics (both resistive and capacitive) into smaller contributions.

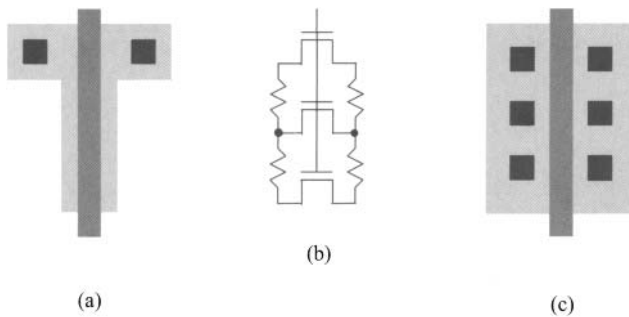


Figure 20.5 (a) Large device with a single contact and (b) its equivalent circuit. (c) Adding more contacts to reduce parasitic resistance.

Examine Fig. 20.6. In this figure, a single device with a large W/L is split into several parallel devices, each with a width one-fourth of the original W . One result of splitting the device into several parts is smaller overall parasitic capacitance associated with the reverse-biased implant substrate diode (the drain or source depletion capacitance to substrate). Since values of C_{db} and C_{sb} are proportional to W , the split device reduces these parasitics by a factor of $(n+1)/2n$ where n is the number of parallel devices and is odd. If n is even, then the C_{sb} is reduced by one-half and C_{db} is reduced by $(n+2)/2n$. As seen in Fig. 20.6b, putting the devices in parallel also reduces the parasitic resistance in series with both the source and the drain (it gets cut roughly in half too).

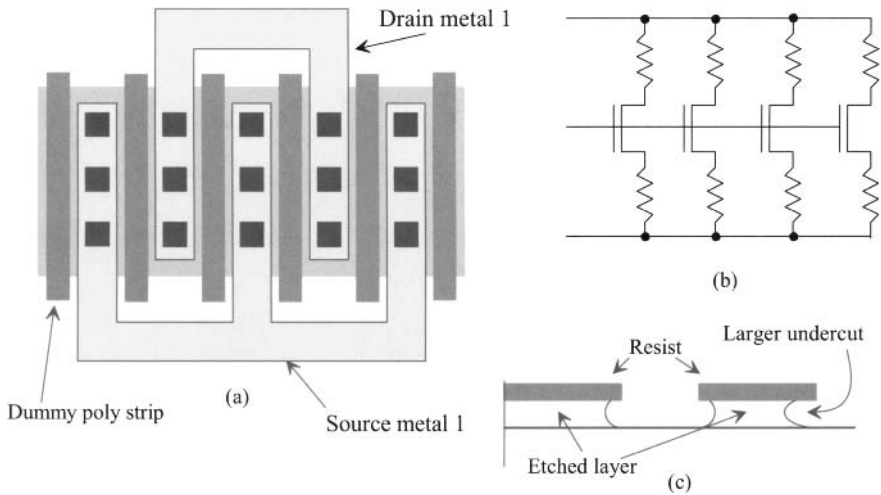


Figure 20.6 (a) A parallel device with dummy strips, (b) the equivalent circuit, and (c) undercutting.

Notice also, that Fig. 20.6a has dummy poly strips on both sides of the device. These strips are used to help minimize the effects of undercutting the poly on the outer edges after patterning, Fig. 20.6c. If the dummy strips had not been used, the poly would have been etched out more under the outermost gates, resulting in a mismatch between the four parallel devices.

When matching two devices, it is imperative that the two devices be as symmetrical as possible. Always orient the two devices in the same direction, unlike those illustrated in Fig. 20.7.

Splitting the devices into parallel devices and interdigitizing them can distribute process gradients across both devices and thus improve matching. An example of this is seen in Fig. 20.8. In (a) the current mirror seen in (b) is laid out. Each MOSFET in (b) is split up into four MOSFETs. If the W/L of each MOSFET in (b) is $80/2$, then the size of each MOSFET (finger) in (a) is $20/2$. Note the use of dummy poly strips on this layout. A good exercise at this point is to lay out the mirror of Fig. 20.8 in a common-centroid arrangement (see Ch. 5). Using a large layout area (long lengths and wide widths) and common-centroid layouts can result in significant improvements in matching.

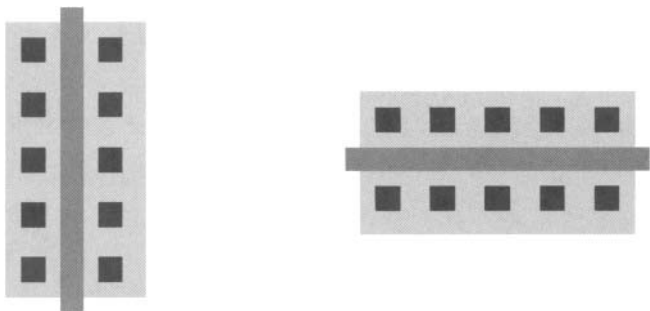


Figure 20.7 Devices with differing orientation (bad).

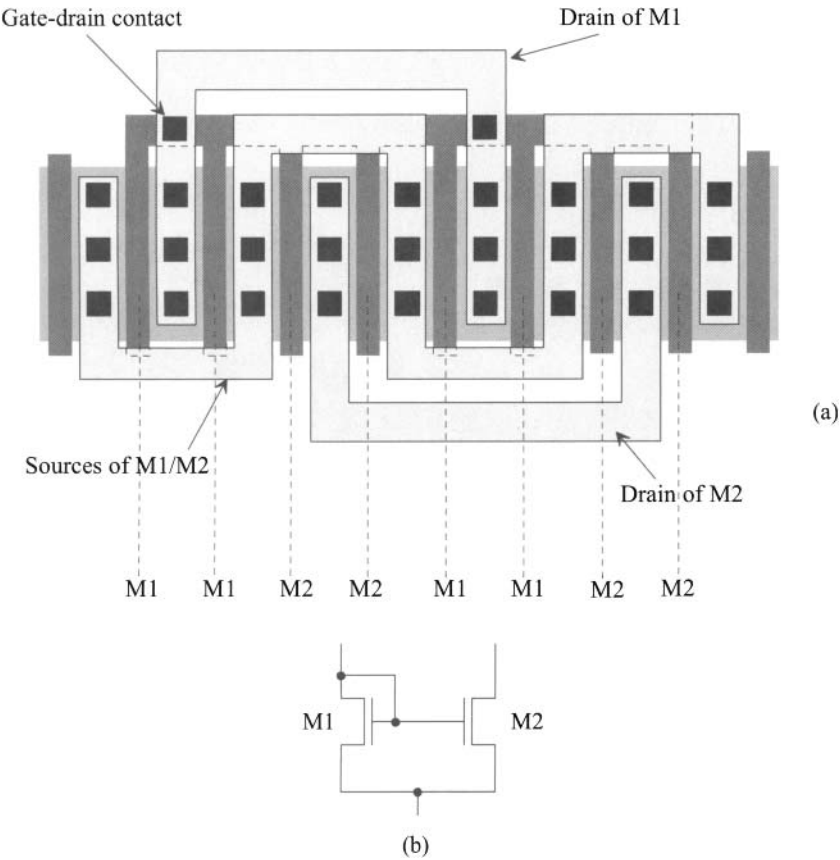


Figure 20.8 (a) Layout of a simple current mirror using interdigitation and (b) equivalent circuit.

Layout of the Mirror with Different Widths

When laying out the current mirror, the lateral diffusion, L_{diff} , under the gate oxide (Fig. 5.13) and the oxide encroachment, W_{enc} , can affect the actual length and width, respectively, of the MOSFET. If not careful with the layout, this will affect the mirror's ratio. Equation 20.3 can be rewritten, without including the differences due to drain-source voltage and lambda, as

$$\frac{I_O}{I_{REF}} = \frac{(W_{2drawn} - 2W_{enc}) \cdot (L_{1drawn} - 2L_{diff})}{(W_{1drawn} - 2W_{enc}) \cdot (L_{2drawn} - 2L_{diff})} \quad (20.11)$$

If the requirement that $L_{1drawn} = L_{2drawn}$ is imposed, then the widths of the devices determine the relative currents in the mirror. Figure 20.9a shows a current mirror layout without width compensation. If W_{enc} is 0.1, then for this layout,

$$\frac{I_O}{I_{REF}} = \frac{40 - .2}{20 - .2} = 2.01 \text{ (a 1\% error due to poor layout)}$$

Figure 20.9b shows how to lay out a current mirror to avoid these problems. The layout of M2 is two MOSFETs in parallel. This can be specified in SPICE by adding $M = X$ after the MOSFET statement in the netlist, where X is the number of MOSFETs,

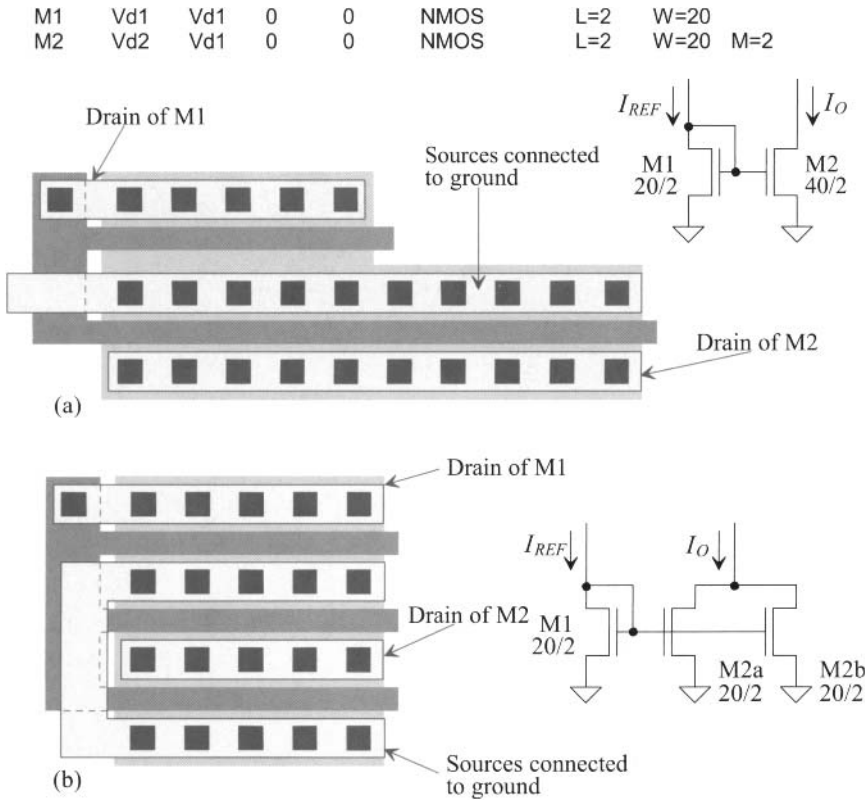


Figure 20.9 Layout of a current mirror (a) without width correction and (b) with width correction.